# Decoding Self-Supervised Learning Embeddings from mIF-H&E Co-registered Melanoma Images: Toward Interpretable Self-Supervised Learning


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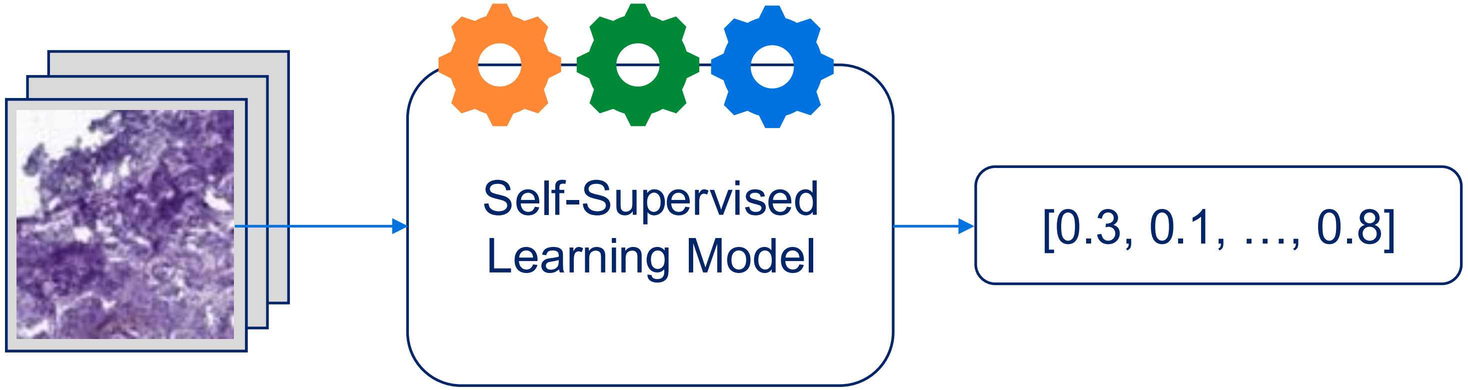
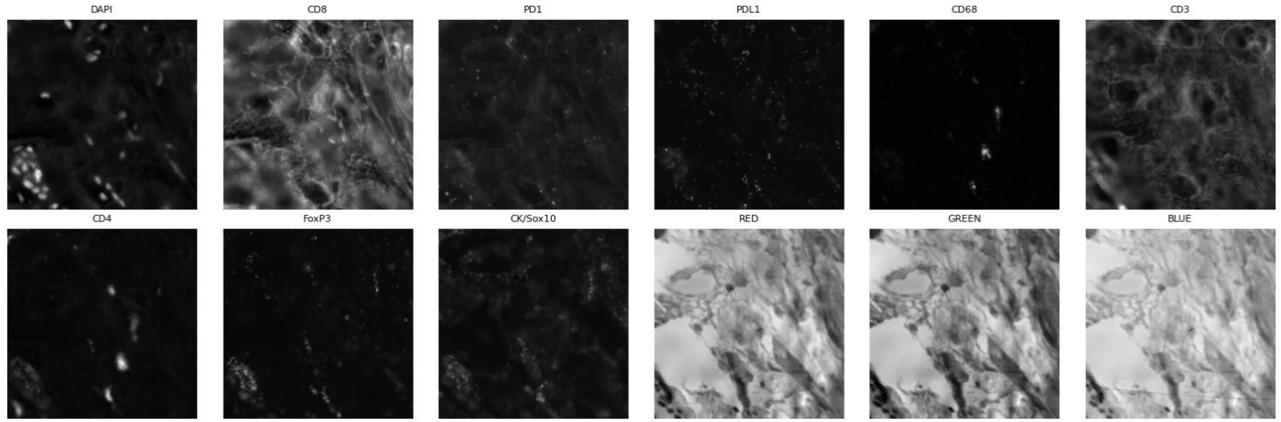




Memorial Sloan Kettering Cancer Center

# Self-Supervised Learning Models Learn Representations of Images, But Remains a Black Box

Our Data: mIF-H&E patches with multiple channels



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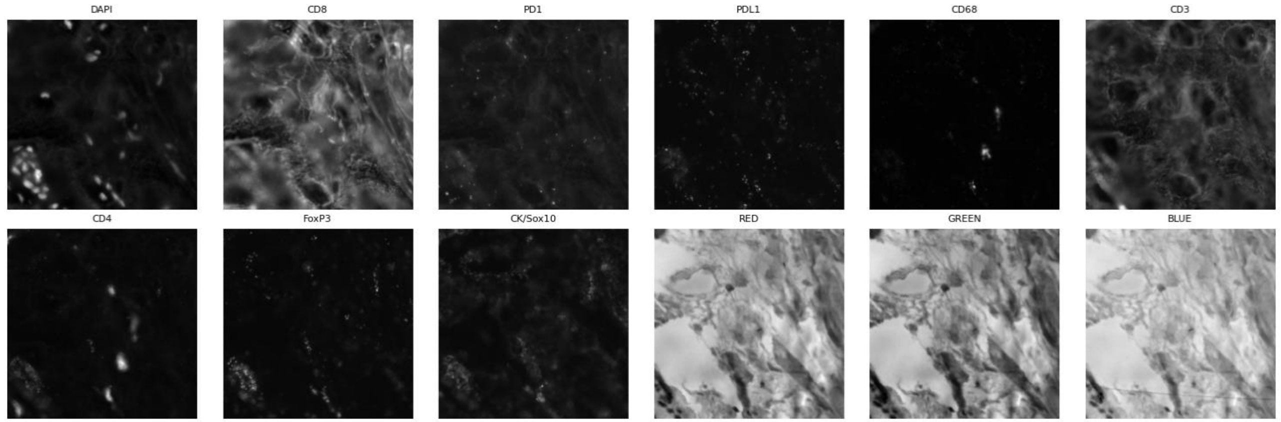


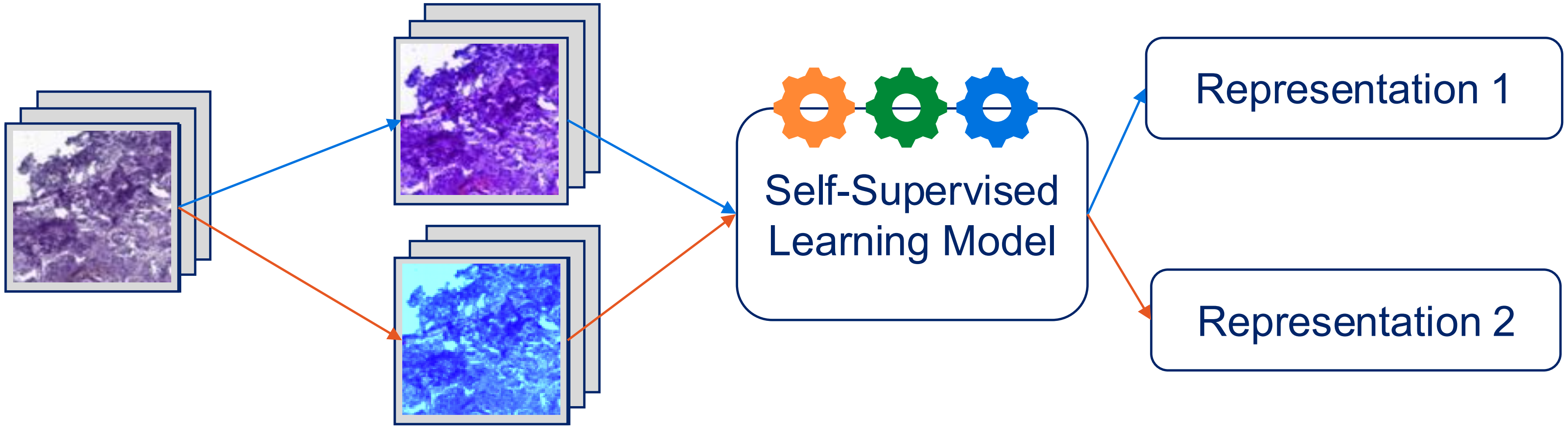


The two representations should be very **similar**!

Each patch has 2 augmented versions.

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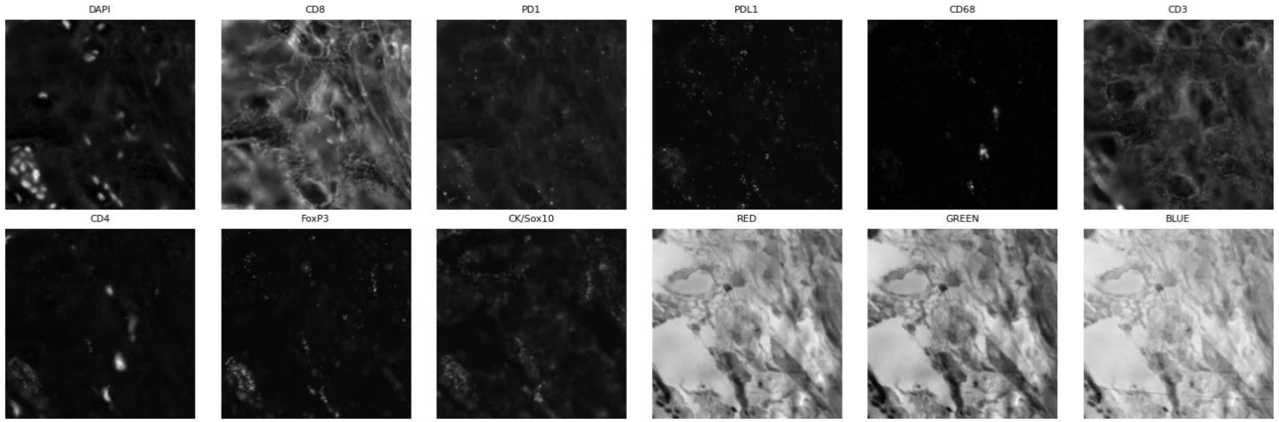
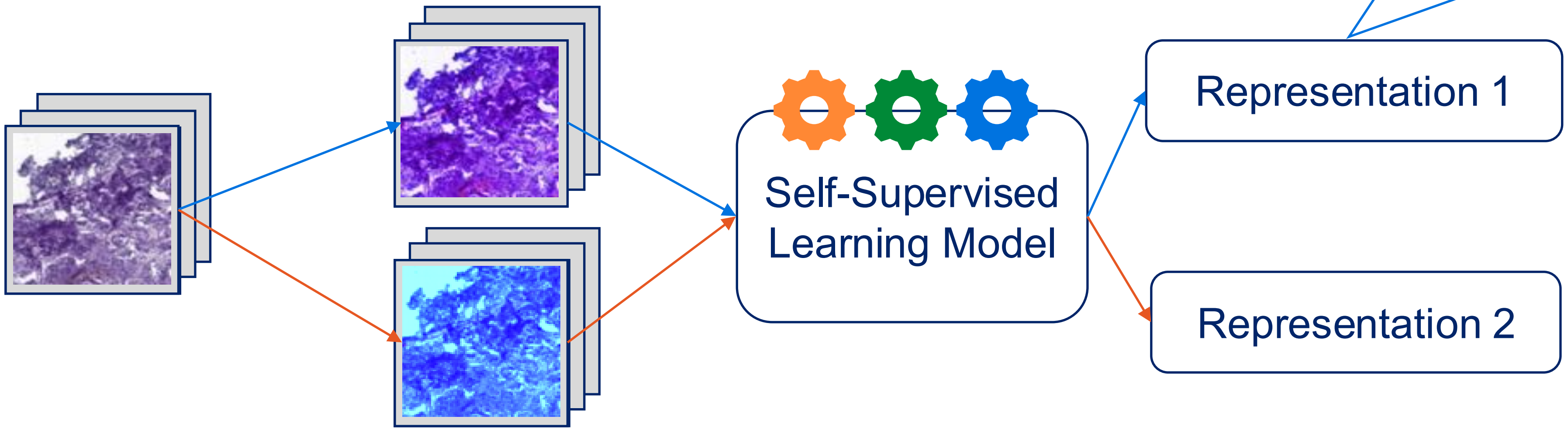



- What is the model learning?
- What is the effect brought by different augmentation choices?

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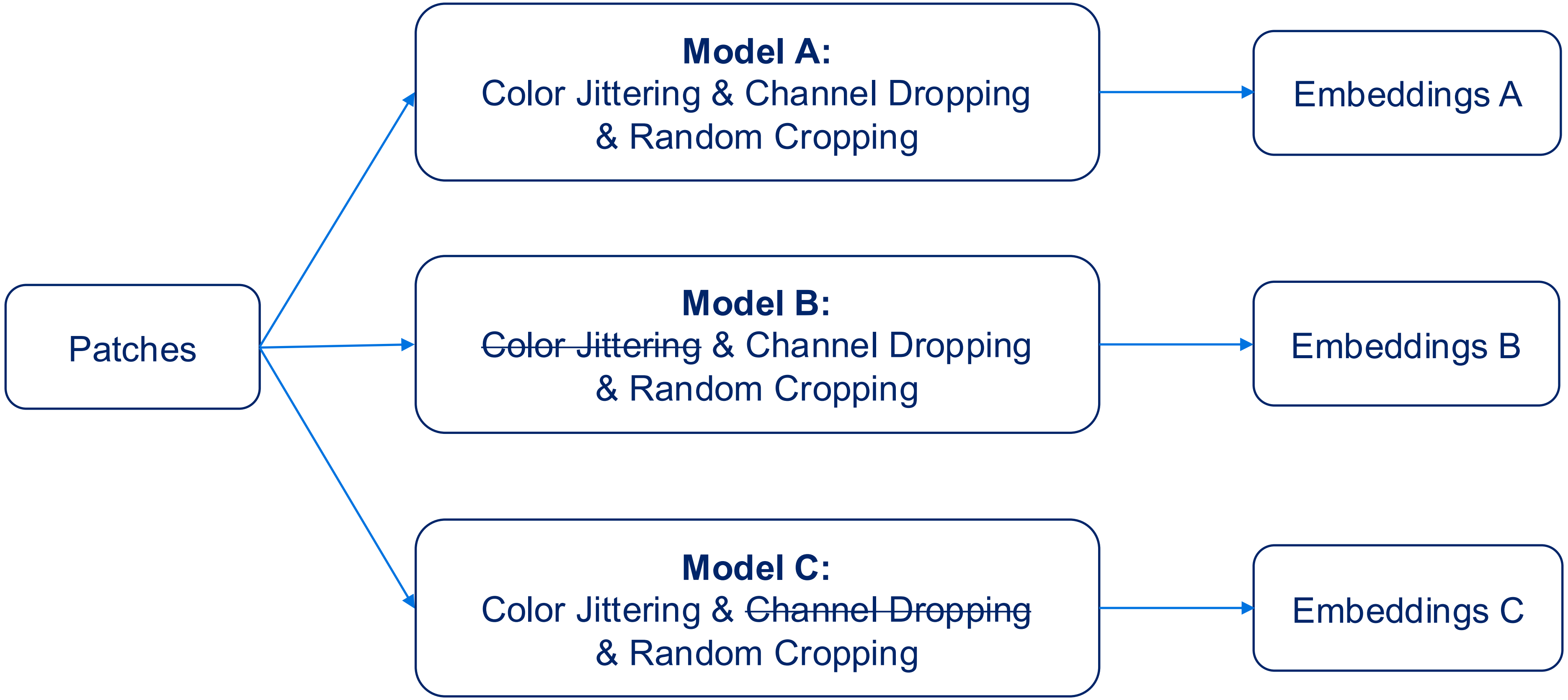
# Model with Different Augmentation Choices were Trained to Investigate the Effect Brought by Different Augmentations




Different augmentation choices → Different Models

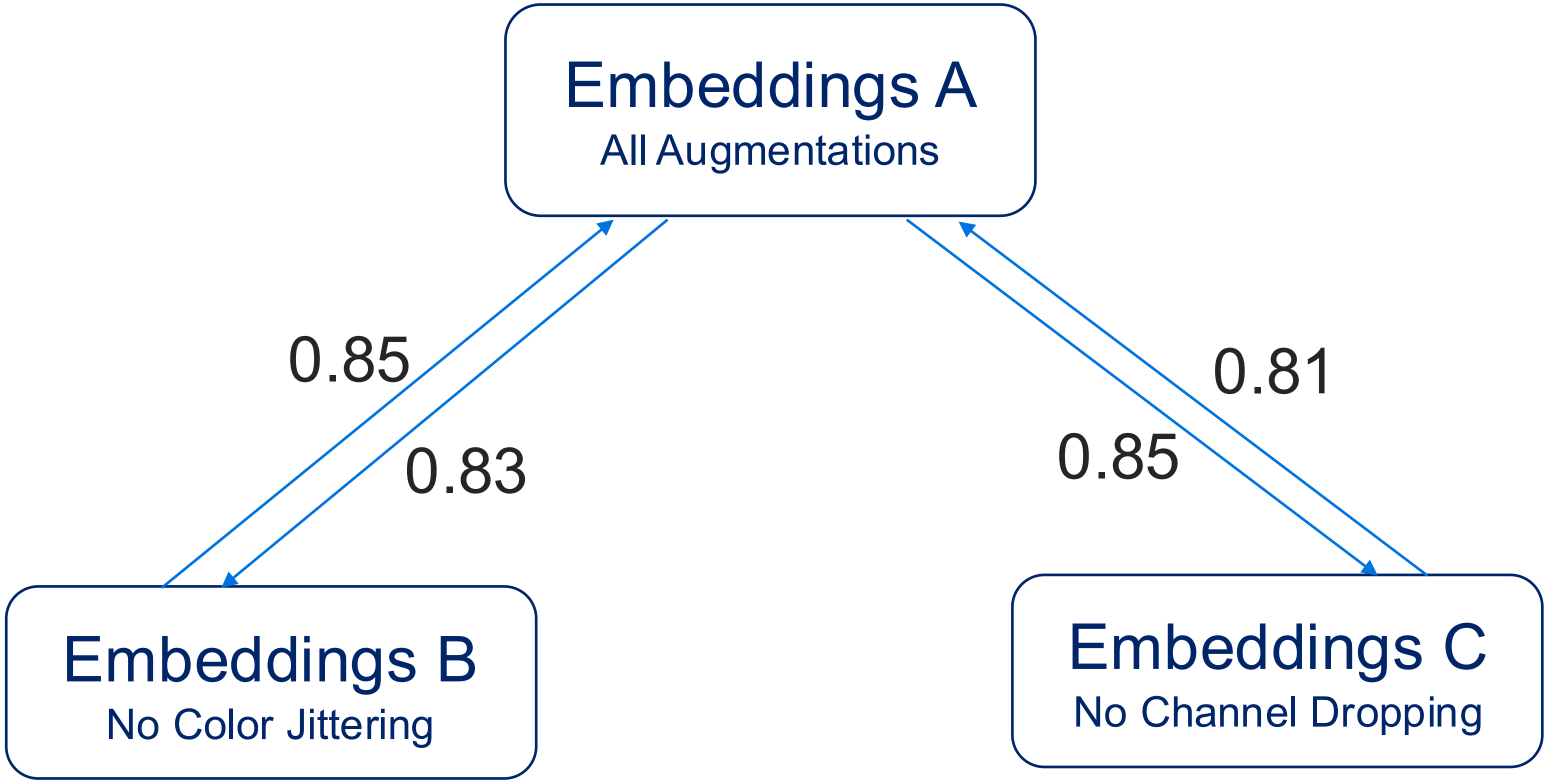
# 3 Questions to Answer

Question 1:
Are the three models capturing the same information?

Question 2:
If not, what information is unique to a specific model?

Question 3:
Is there specific features in each model that are responsible for the difference?

# Reciprocal Prediction Test Reveals Differences in the Information Learned in Models w/ Different Augmentations




Random Forest Prediction Performance (validation $R^2$)

Most of the information captured by the models is shared, with each model capturing a small amount of unique information.

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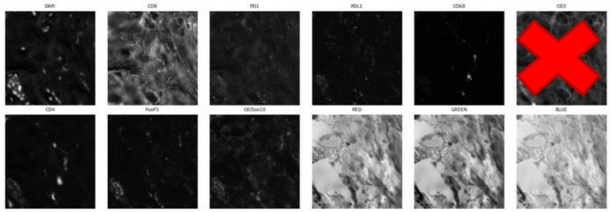
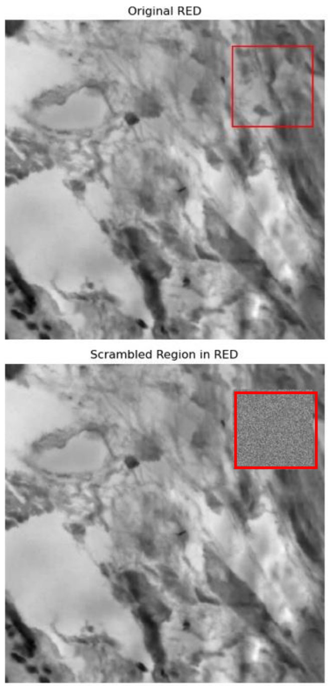
# Controlled Experiments to Identify Model-Level Properties

## Integrating Information across Multiple Channels



Replace it with the channel from another patch.

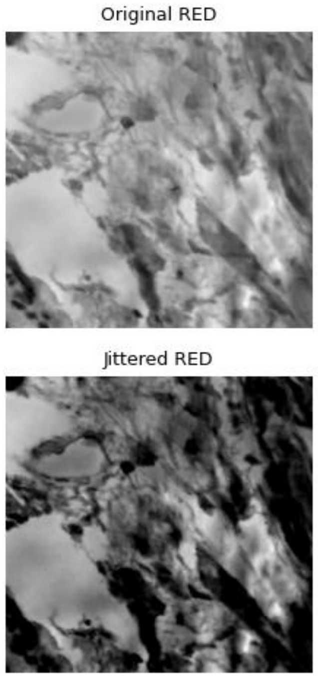
Different Embeddings?




## Encoding Spatial Structure

Different Embeddings?

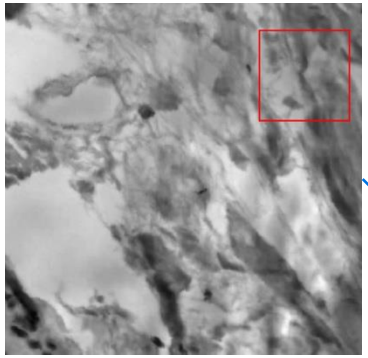
Actual scrambling region smaller than shown.



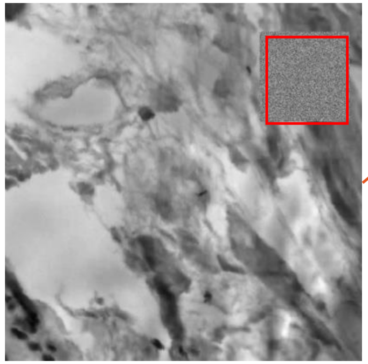

## Stable Under Color Variance

Similar Embeddings?

# Controlled Experiments to Identify Model-Level Properties
# Step 1: Prepare Datasets



Not scrambled



Scrambled

Self-Supervised Learning Model

| $z_1$ | … | $z_{512}$ | Scramble |
|---|---|---|---|
| 0.00 | … | 0.03 | 0 = No |
| 0.05 | … | 0.01 | 1 = Yes |
| … | … | … | … |

Repeat this for many patches to get the complete dataset.

# Controlled Experiments to Identify Model-Level Properties
# Step 2: Predict Scrambling using Embeddings from three models

Train Random Forest Classifier.

| $z_1$ | … | $z_{512}$ | Scramble |
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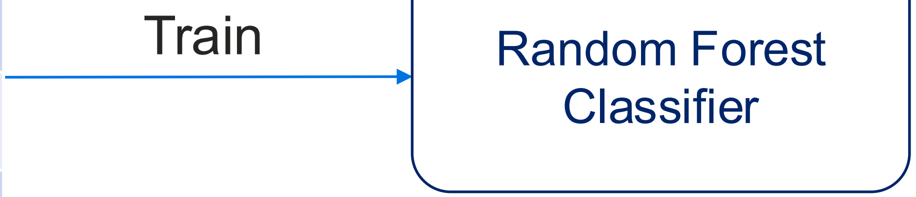
Random Forest Classifier Input  Target of prediction



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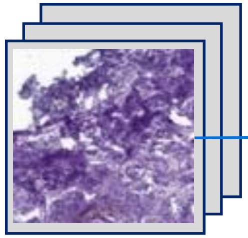
Random Forest Classifier Input — Target of prediction

Train → Random Forest Classifier

Gather performance from validation set.



Unseen patch
Scrambled / Not

Self-Supervised Learning Model

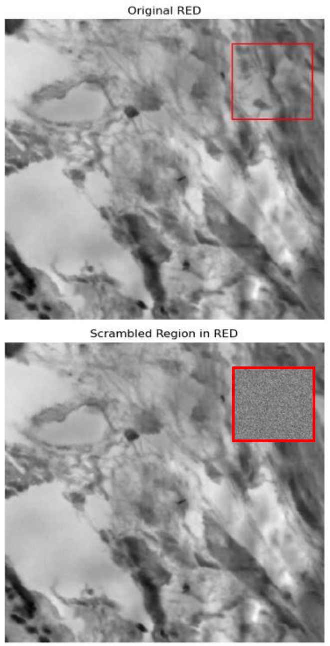
| $z_1$ | … | $z_{512}$ |
|---|---|---|
| 0.01 | … | 0.07 |

Random Forest Classifier

Correct Prediction

Incorrect Prediction

# Embeddings From the Model with Both Augmentation Best Predicts Spatial Scrambling

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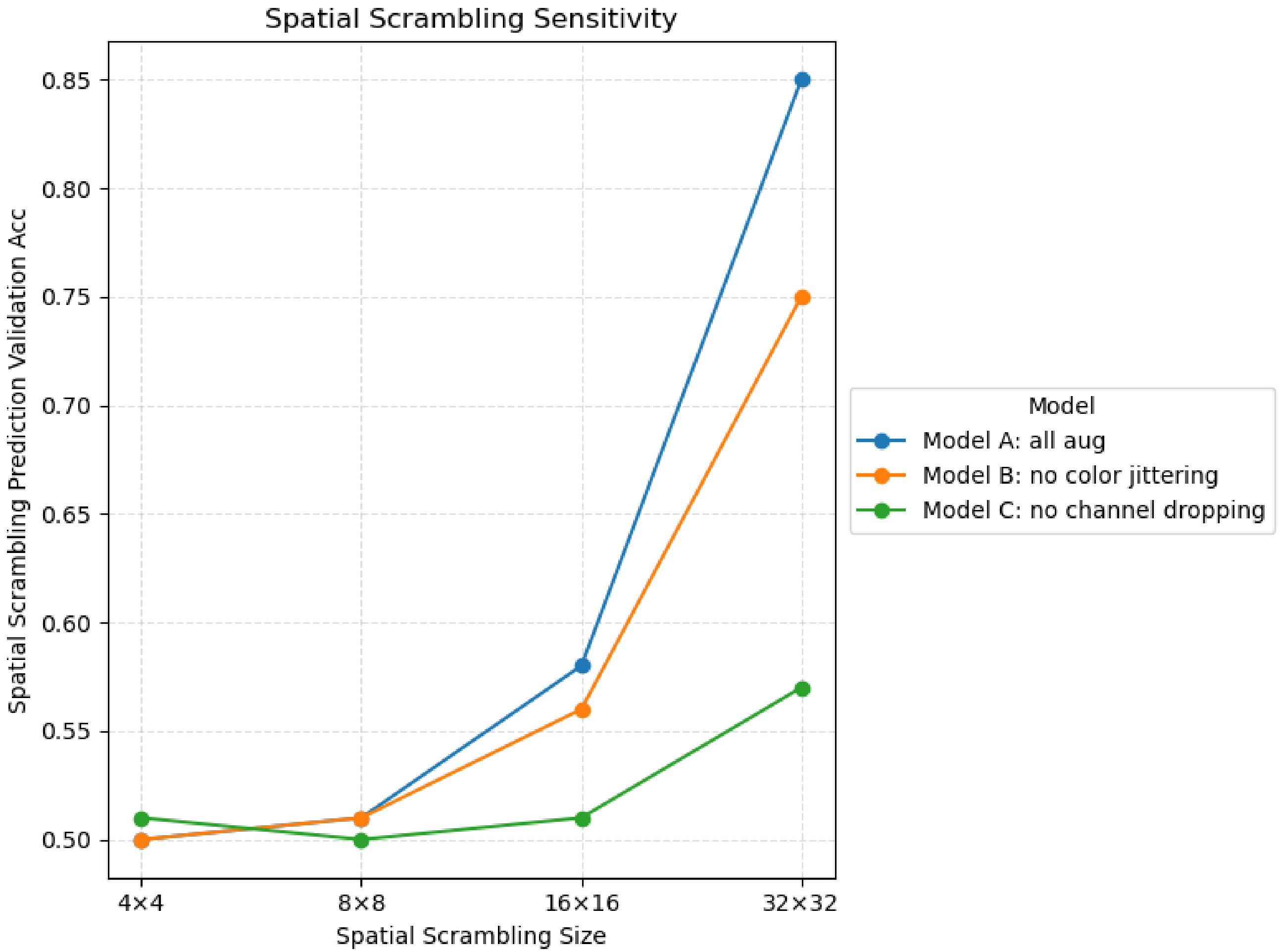


Baseline acc for task: 0.50

# 3 Questions to Answer

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Model w/ all augmentations encodes unique spatial information.

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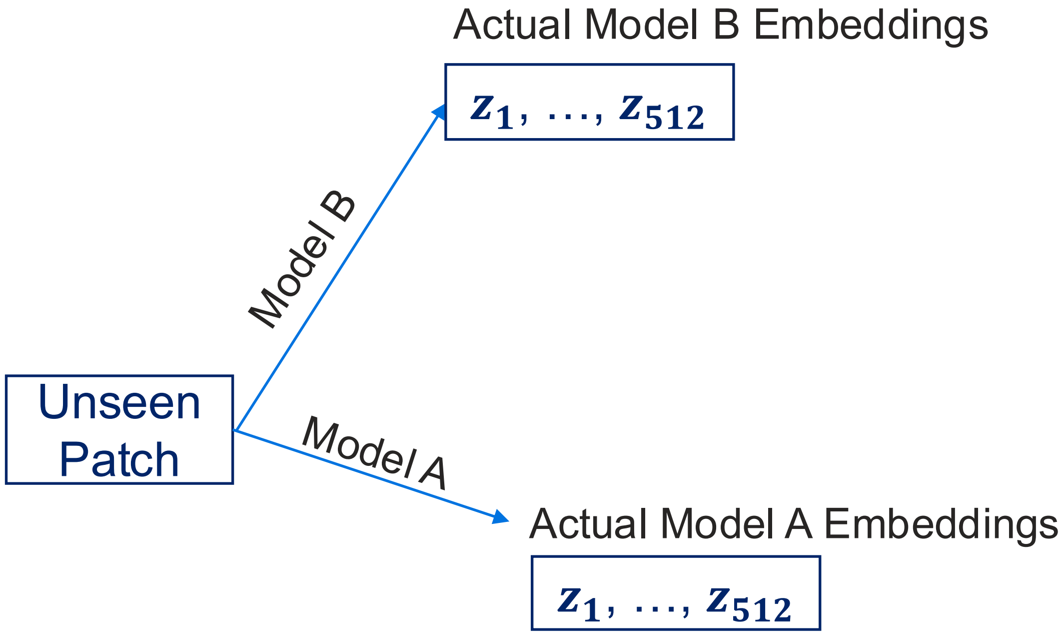
Is there specific features in each model that are responsible for the difference?

# Feature-wise Comparison between True and Predicted Embeddings Also Reveals the Differences in Information Learned

▲ = Trained Random Forest Regressor
Take Embeddings B to predict Embeddings A



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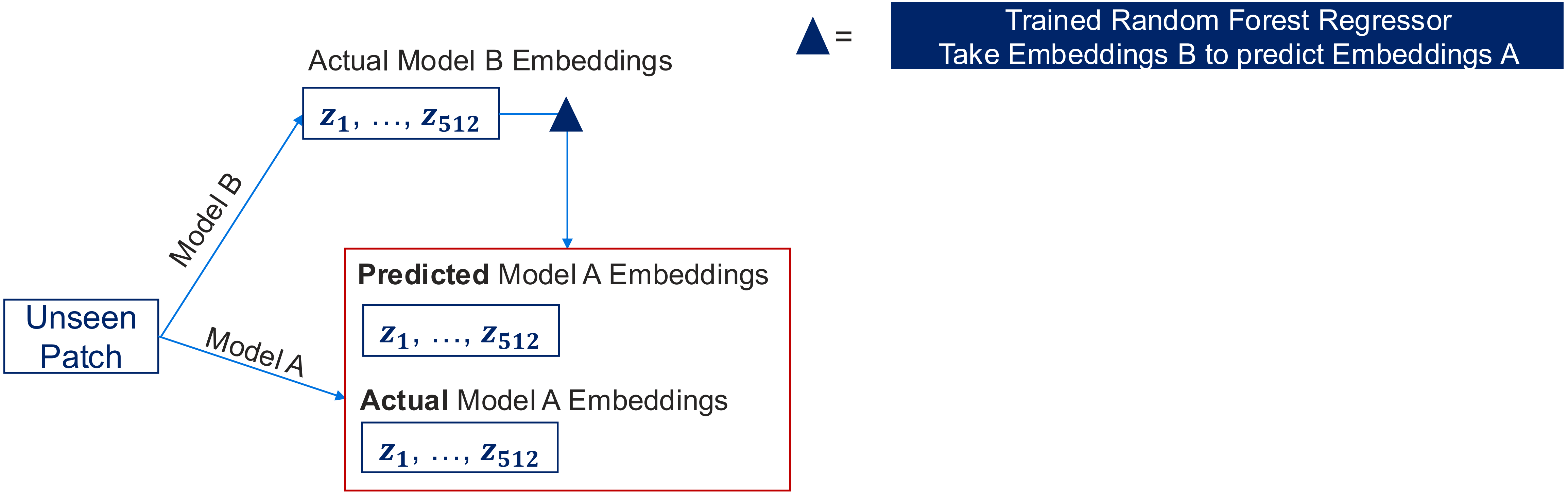

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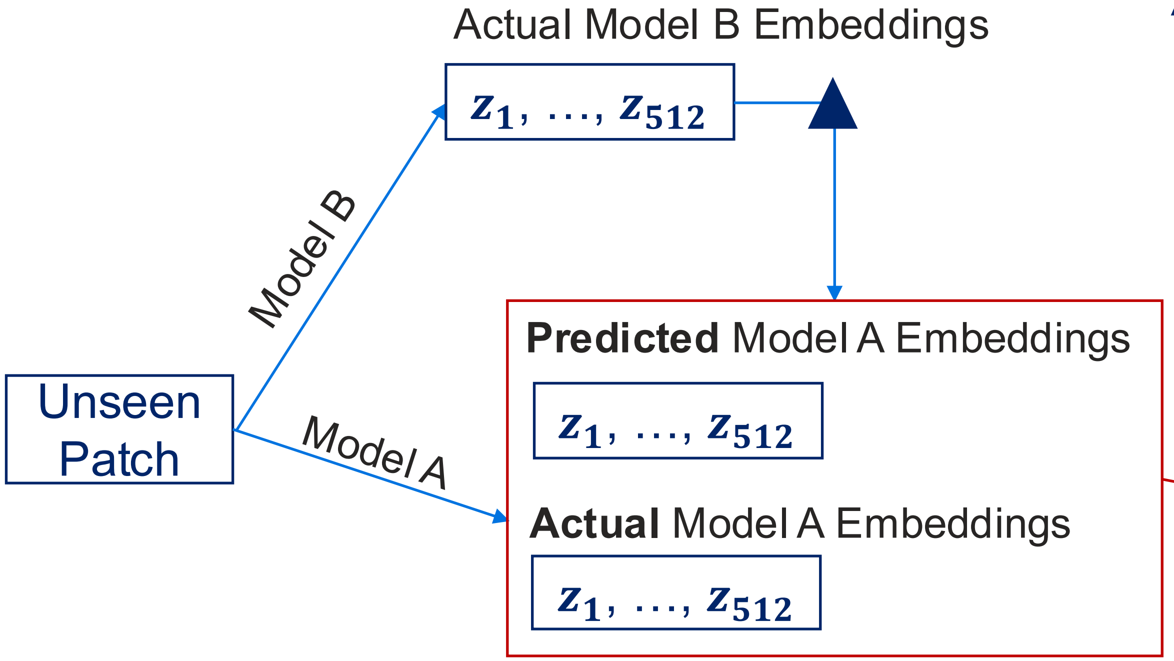


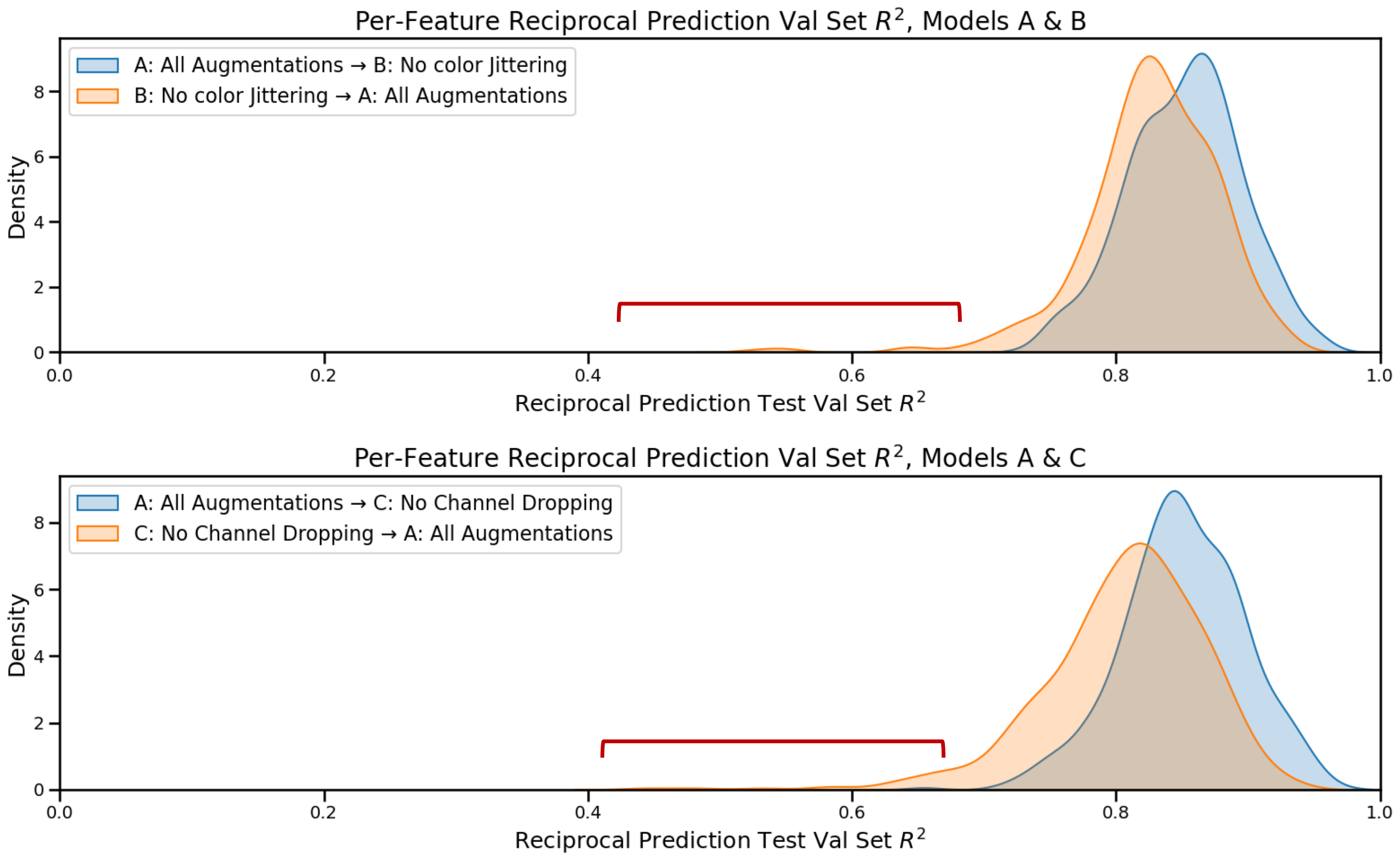

Column-wise Comparison:
How well is each feature predicted?

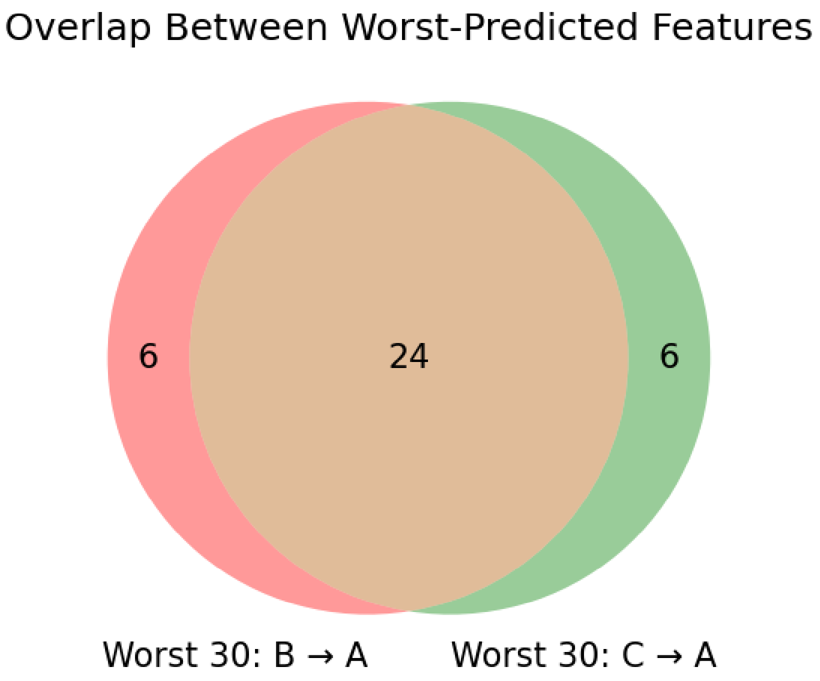
| **Actual** | $z_1$ | … | $z_{512}$ |
|---|---|---|---|
| Patch 1 | | | |
| … | | | |
| Patch N | | | |

| **Predicted** | $z_1$ | … | $z_{512}$ |
|---|---|---|---|
| Patch 1 | | | |
| … | | | |
| Patch N | | | |

Now we know, for each feature:
How well is it predicted B→A and C→A?

# A Set of Model A Features Can't Be Reconstructed from Model B&C.

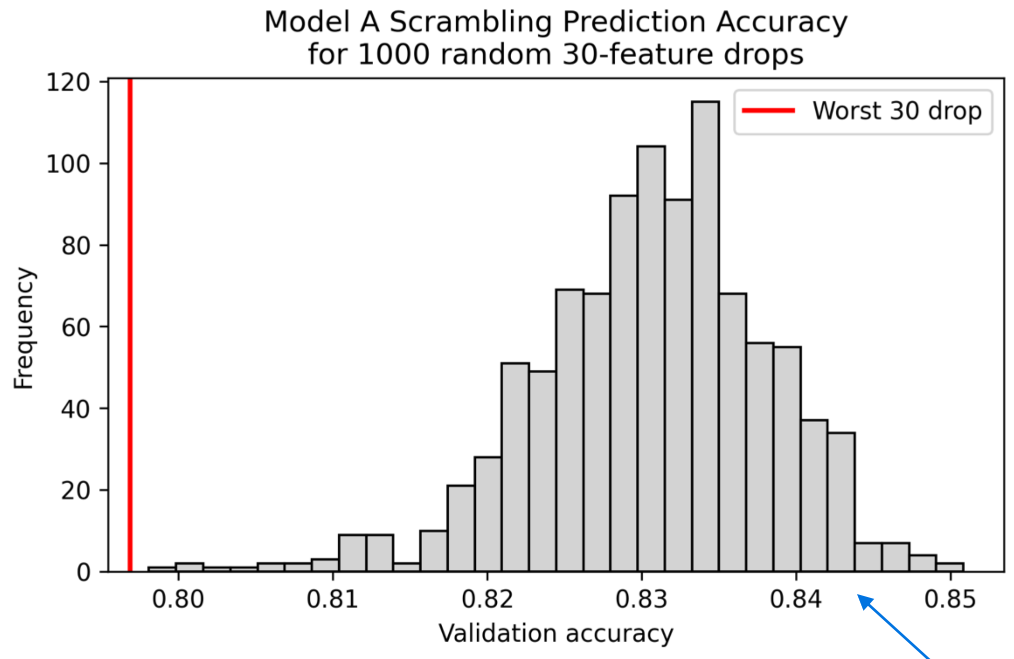



**The long tail = Features that are poorly predicted from other two models to model w/ all aug.**
Are they what makes the model with all augmentations better at detecting spatial scrambling?

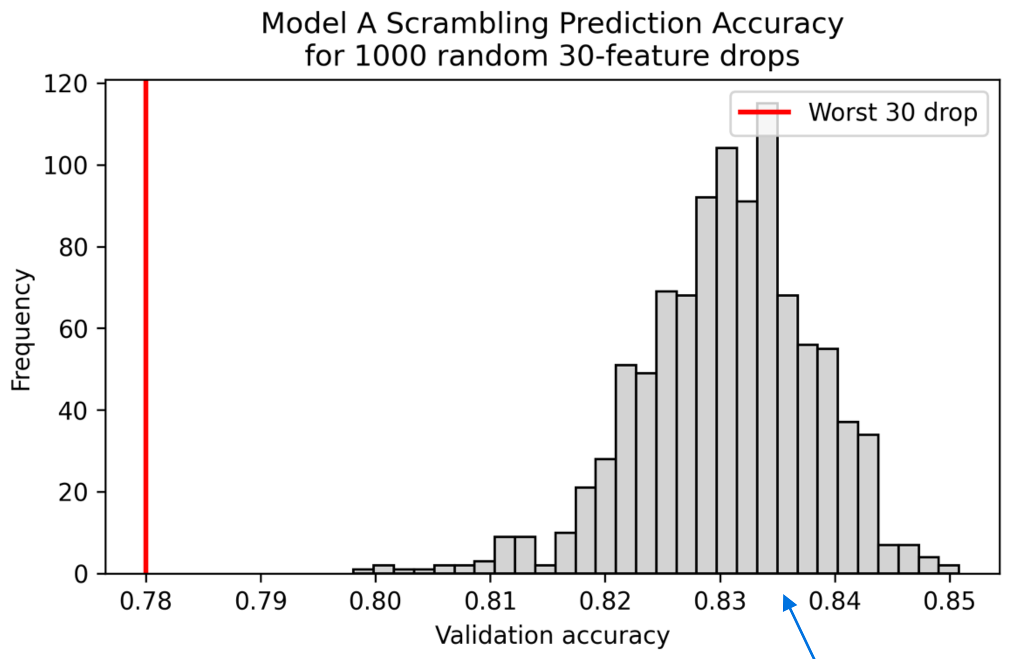
# Removing Such a Set of Features Decreases Model A's Ability in Detecting Spatial Scrambling

Remove 30 worst predicted features, B→A




Monte Carlo p-value: 0.001

Remove 30 worst predicted features, C→A




Monte Carlo p-value: 0.001

Randomly remove 30 features (in 512 features) 1000 times to generate an accuracy distribution for predicting spatial scrambling (32*32 pixels).

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Model with all augmentations encodes unique spatial information.

Yes, but in an ambiguous way.

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# Conclusions

Different augmentations → Different information captured.

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# Conclusions

Different augmentations → Different information captured.

Model w/ color-jitter + channel-drop → Most Sensitive to Spatial Structure

Yes, but in an ambiguous way.

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Is there specific features in each model that are responsible for the difference?

# Conclusions

Different augmentations → Different information captured.

Model w/ color-jitter + channel-drop → Most Sensitive to Spatial Structure

Spatial Information: encoded in many features, while worst predicted features contain more spatial information.

# Next Steps

- More well-rounded controlled experiments.
- Explore more different settings during the training process:
  - Effect of different embedding sizes.
  - Different extents of color jittering / other augmentations.
- Relationship between features: better ways to model how multiple pieces of information overloads a feature and how multiple features are correlated.
  - Possible approach: Networks

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